

1. Personal Insight

One of the most interesting aspects of this lab was working with an extremely imbalanced dataset. This challenged my assumptions about model performance, especially seeing how misleading accuracy can be in such contexts. The most difficult part was understanding how to properly evaluate the model using precision, recall, and F1-score instead of relying on accuracy alone. IBM Watson Studio was also a notable experience—it felt more structured and scalable compared to local development, but slightly more complex due to cloud setup, tokens, and data access steps. This lab clearly demonstrated how fraud detection is a real-world cybersecurity problem, where models must operate quickly and adapt to constantly evolving threats.

2. Technical Analysis

The Logistic Regression model provided a solid baseline, but its performance on fraud detection was limited, particularly in recall. In contrast, the Random Forest model with class weighting significantly improved the ability to detect fraudulent transactions, resulting in a higher F1 score. This highlights why accuracy is a poor metric in imbalanced datasets. It can appear very high simply by predicting most of the class. In a production system, I would likely deploy the Random Forest model because it better balances precision and recall. However, there are trade-offs: false positives can inconvenience users, while false negatives allow fraud to occur. In financial systems, minimizing false negatives (catching fraud) is typically prioritized, even if it increases false positives.

3. Extension Ideas

To extend this lab, I would experiment with SMOTE to address class imbalance or try anomaly detection methods like Isolation Forest. Additionally, implementing real-time fraud detection of API or exploring deep learning models could make the system more production-ready and adaptive.